Multivariate Data Analysis - MESIO UPC

Principal Component Analysis

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A bit of history

- Pearson, K. (1901) *On lines and planes of closest fit to systems of points in space* Philosophical Magazine 6(2): 559-572.
- Hotelling, H. (1933) *Analysis of a complex of statistical variables into principal components*, Journal of Educational Psychology, 24: 417-441,498-520.
- Gabriel, K. R. (1971) *The biplot graphic display of matrices with application to principal component analysis*, Biometrika, 58(3): 453-467.

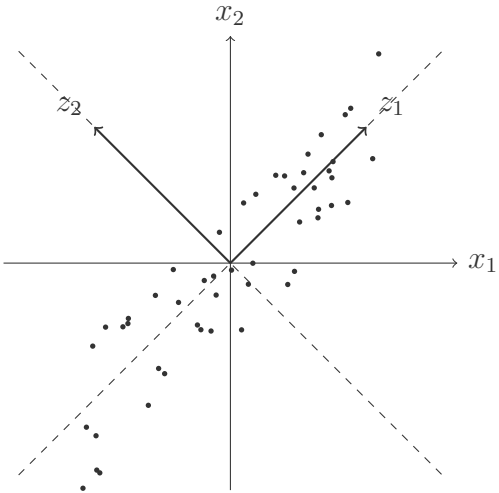
PCA

Principal **C**omponent **A**nalysis is a multivariate tool used to explain the variance-covariance structure of a set of variables through as few linear combinations as possible. The new (artificial) variables are called **Principal Components**.

The goals of the analysis are:

- **Extract** information regarding the relationship between variables and observations.
- **Reduce** the dimensionality of the dataset while retaining as much variance (information) as possible.
- **Visualize** the data through a **biplot** so observations and variables can be related.
- **Identify** latent (hidden) structures and patterns within the data that are not immediately apparent.

PCA



Theory of PCA (1): Context

We will use the following components:

- $X_1, \ldots, X_p$ random variables.
- $\boldsymbol{\Sigma}$ the covariance matrix.
- $\boldsymbol{\rho}$ the correlation matrix.

No assumption about distribution of the random variables $X_1, \ldots, X_p$ but multivariate normal has some nice properties and inferences.

Theory of PCA (2): Framework (1)

Having a random vector $\mathbf{X}_{p\times 1}$ with covariance matrix $\mathbf{\Sigma}$. We consider:

$$Z_i = \mathbf{a}_i^t \mathbf{X} = a_{i1}X_1 + a_{i2}X_2 + \cdots + a_{ip}X_p$$

To be the *i*th linear combination of the r.v. $X_1, \ldots X_p$ for $i = 1, \ldots, p$. These new random variables have the following characteristics:

- $Var(Z_i) = a_i^t \mathbf{\Sigma} a_i$ for $i = 1, \ldots, p$. **Large variance**.
- $Cov(Z_i, Z_j) = a_i^t \mathbf{\Sigma} a_j = 0$ for $i, j = 1, \ldots, p$ and $i \neq j$. **Uncorrelated**.

The Z_i variables will be the **Principal Components**. We want them uncorrelated and with variance as large as possible.

Theory of PCA (3): Framework (2)

$$
\begin{aligned}
Z_1 &= a_{11}X_1 + a_{12}X_2 + \cdots + a_{1p}X_p \\
Z_2 &= a_{21}X_1 + a_{22}X_2 + \cdots + a_{2p}X_p \\
&\vdots \\
Z_p &= a_{p1}X_1 + a_{p2}X_2 + \cdots + a_{pp}X_p
\end{aligned}
$$

subject to:

- $Z_1, \ldots, Z_p$ uncorrelated; $\min(n-1, p)$ components.
- $Var(Z_1)$ to be maximal.
- $Var(Z_1) \geq Var(Z_2) \geq \cdots \geq Var(Z_p)$
- $\|\mathbf{a}_i\| = 1$

Theory of PCA (3): Optimization problem (1)

- Let $\mathbf{X}$ be a $p \times 1$ centered random vector with covariance $\mathbf{\Sigma}$ and Z_1 the linear combination given by $\mathbf{a}_1^t \mathbf{X}$.
- Maximize the variance of the first component

$$V(Z_1) = \mathbf{a}_1^t \mathbb{E}[\mathbf{X}\mathbf{X}^t]\mathbf{a}_1 = \mathbf{a}_1^t \mathbf{\Sigma} \mathbf{a}_1$$

with restriction $\mathbf{a}_1^t \mathbf{a}_1 = 1$.

Theory of PCA (3): Optimization problem (2)

- Obtain the Lagrangian:

$$L(\mathbf{a}_1, \lambda_1) = \mathbf{a}_1^t \boldsymbol{\Sigma} \mathbf{a}_1 - \lambda_1(\mathbf{a}_1^t \mathbf{a}_1 - 1)$$

  Derive with respect to $\mathbf{a}_1$ and λ_1:

$$\frac{\partial L}{\partial \mathbf{a}_1} = 2\boldsymbol{\Sigma}\mathbf{a}_1 - 2\lambda_1 \mathbf{a}_1; \qquad \frac{\partial L}{\partial \lambda_1} = 1 - \mathbf{a}_1^t \mathbf{a}_1$$

- Set first order derivatives to zero

$$\boldsymbol{\Sigma}\mathbf{a}_1 = \lambda_1 \mathbf{a}_1; \qquad \mathbf{a}_1^t \mathbf{a}_1 = 1 \tag{1}$$

- λ_1 and $\mathbf{a}_1$ are eigenvalue and eigenvector of $\boldsymbol{\Sigma}$, and $\mathbf{a}_1$ has unit length.

Theory of PCA (3): Optimization problem (*)

Observe that pre-multiplication of (1) by $\mathbf{a}_1^t$ gives:

$$\mathbf{a}_1^t \boldsymbol{\Sigma} \mathbf{a}_1 = \mathbf{a}_1^t \lambda_1 \mathbf{a}_1 \iff$$
$$Var(Z_1) = \lambda_1 \mathbf{a}_1^t \mathbf{a}_1 \iff$$
$$Var(Z_1) = \lambda_1$$

λ_1 is the **variance** of the first principal component.

Important

$Z_1 = \mathbf{a}_1^t \mathbf{X}$: First Principal Component, and $\mathbf{a}_1$ is the first eigenvector of $\boldsymbol{\Sigma}$. $Var(Z_1) = \lambda_1$ where λ_1 is the first eigenvalue of $\boldsymbol{\Sigma}$.

Theory of PCA (3): Optimization problem (3)

We now repeat the process to identify the second component, with the added constraint that it must be orthogonal to the previous ones.

- The second component ($Z_2 = \mathbf{a}_2^t \mathbf{X}$) without correlation (orthogonal) to the first, has variance

$$V(Z_2) = \mathbf{a}_2^t \mathbb{E}[\mathbf{X}\mathbf{X}^t]\mathbf{a}_2 = \mathbf{a}_2^t \boldsymbol{\Sigma} \mathbf{a}_2$$

  to be maximized subject to $\mathbf{a}_2^t \mathbf{a}_2 = 1$ and $\mathbf{a}_1^t \boldsymbol{\Sigma} \mathbf{a}_2 = 0$

Theory of PCA (3): Optimization problem (4)

- The Lagrangian is:

$$L(\mathbf{a}_2, \lambda_2, \mu) = \mathbf{a}_2^t \boldsymbol{\Sigma} \mathbf{a}_2 - \lambda_2(\mathbf{a}_2^t \mathbf{a}_2 - 1) - \mu(\mathbf{a}_1^t \boldsymbol{\Sigma} \mathbf{a}_2)$$

Setting first order derivatives equal to zero:

$$\frac{\partial L}{\partial \mathbf{a}_2} = 2\boldsymbol{\Sigma}\mathbf{a}_2 - 2\lambda_2 \mathbf{a}_2 - \mu \boldsymbol{\Sigma}\mathbf{a}_1 = 0 \tag{2}$$

$$\frac{\partial L}{\partial \lambda_2} = 1 - \mathbf{a}_2^t \mathbf{a}_2 = 0, \qquad \frac{\partial L}{\partial \mu} = \mathbf{a}_1^t \boldsymbol{\Sigma} \mathbf{a}_2 = 0$$

Theory of PCA (3): Optimization problem (*)

Observe that pre-multiplication of (2) by $\mathbf{a}_2^t$ gives:

$$
\begin{aligned}
2\mathbf{a}_2^t\boldsymbol{\Sigma}\mathbf{a}_2 - 2\lambda_2\mathbf{a}_2^t\mathbf{a}_2 - \mu\mathbf{a}_2^t\boldsymbol{\Sigma}\mathbf{a}_1 = 0 &\iff \\
2\mathbf{a}_2^t\boldsymbol{\Sigma}\mathbf{a}_2 = 2\lambda_2\mathbf{a}_2^t\mathbf{a}_2 + \mu\mathbf{a}_2^t\boldsymbol{\Sigma}\mathbf{a}_1 &\iff \\
Var(Z_2) = \lambda_2\mathbf{a}_2^t\mathbf{a}_2 &\iff \\
Var(Z_2) = \lambda_2 &
\end{aligned}
$$

λ_2 is the **variance** of the second principal component subject to constraints $\mathbf{a}_2^t\mathbf{a}_2 = 1$ and $\mathbf{a}_2^t\boldsymbol{\Sigma}\mathbf{a}_1 = 0$. These ensure the second eigenvector has unit length and remains orthogonal to the first.

Important

$Z_2 = \mathbf{a}_2^t\mathbf{X}$: Second Principal Component, and $\mathbf{a}_2$ is the second eigenvector of $\boldsymbol{\Sigma}$. $Var(Z_2) = \lambda_2$ where λ_2 is the second eigenvalue of $\boldsymbol{\Sigma}$.

Theory of PCA (3): Optimization problem (6)

Note

Continue the process to obtain the ith eigenvector of the covariance matrix; its corresponding eigenvalue λ_i, represents the variance captured by that component.

Theory of PCA (3): Optimization problem (7)

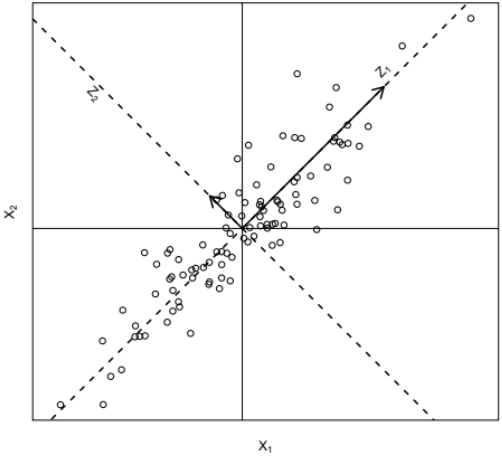
- $\mathbf{a}_i$: Is the vector of coefficients in the linear combination $Z_i = \mathbf{a}_i^t X$. It is the i-th eigenvector of $\mathbf{\Sigma}$ and in the context of PCA, these coefficients are formally referred to as **loadings**.

- All loadings and eigenvalues can be obtained by the spectral decomposition of the covariance matrix:

$$\mathbf{\Sigma} = \mathbf{V}\mathbf{D}_\lambda\mathbf{V}^t.$$

- Consequently, $\mathbf{V} = \left(\mathbf{a}_1|\mathbf{a}_2|\ldots|\mathbf{a}_p\right)$

Geometric Interpretation ($p = 2$)



Theory of PCA (4): Outcomes (1)

Let $\mathbf{\Sigma}$ be the covariance matrix associated with the random vector $\mathbf{X}^t = (X_1, X_2, \ldots, X_p)$. Let $\mathbf{\Sigma}$ have the eigenvalue-eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2) \ldots (\lambda_p, \mathbf{e}_p)$ where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$. Then the ith principal component is given by

$$Z_i = \mathbf{e}_i^t \mathbf{X} = e_{i1} X_1 + e_{i2} X_2 + \cdots + e_{ip} X_p \qquad i = 1, 2, \ldots, p$$

With conditions:

$$Var(Z_i) = \mathbf{e_i^t \Sigma e_i} = \lambda_i \qquad i = 1, 2, \ldots, p$$
$$Cov(Z_i, Z_j) = \mathbf{e_i^t \Sigma e_j} = \mathbf{0} \qquad i \neq j$$

Theory of PCA (4): Outcomes (2)

Let $\mathbf{X}^t = (X_1, X_2, \ldots, X_p)$ be a random vector with covariance matrix $\mathbf{\Sigma}$. Let the eigenvalue-eigenvector pairs of $\mathbf{\Sigma}$ be $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \ldots, (\lambda_p, \mathbf{e}_p)$ where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$. Let $Z_i = \mathbf{e}_i^t \mathbf{X} = e_{i1}X_1 + e_{i2}X_2 + \cdots + e_{ip}X_p$ with $i = 1, 2, \ldots, p$ be the principal components. The i-th principal component is defined as the linear combination:

$$\sum_{i=1}^{p} Var(X_i) = tr(\mathbf{\Sigma}) \iff$$

$$tr(\mathbf{\Sigma}) = \sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp} \iff$$

$$tr(\mathbf{\Sigma}) = tr(\mathbf{V}\mathbf{D}_\lambda\mathbf{V}^t) \iff$$

$$tr(\mathbf{\Sigma}) = tr(\mathbf{D}_\lambda) \iff$$

$$tr(\mathbf{D}_\lambda) = \lambda_1 + \lambda_2 + \cdots + \lambda_p \implies$$

$$\sum_{i=1}^{p} Var(X_i) = \sum_{i=1}^{p} Var(Z_i)$$

Theory of PCA (4): Outcomes (2b)

Important

As a consequence, the **proportion of total population variance due to *k*th principal component** is given by:

$$\frac{\lambda_k}{\sum_i \lambda_i} \qquad k = 1, 2, \ldots, p$$

Theory of PCA (4): Outcomes (3)

Let $Z_1, Z_2, \ldots, Z_p$ be the principal components derived from the covariance matrix $\boldsymbol{\Sigma}$, where $Z_i = \mathbf{e_i^t} \boldsymbol{X}$. The **correlation** coefficient between the i-th principal component Z_i and the k-th original variable X_k is given by:

$$\rho_{Z_i, X_k} = e_{ik} \frac{\sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}} \qquad i, k = 1, 2, \ldots, p$$

These correlations, called **factor loadings** in the PCA literature, quantify the linear relationship between the components and the variables. The **factor loadings** provides how much information (variance) from X_k is explained by Z_i

Theory of PCA (4): Outcomes (4)

One can also evaluate the relative contribution of a variable to a component by examining the magnitude of the eigenvector entry $\mathbf{e}_{ik}$ directly. A large absolute value $|\mathbf{e}_{ik}|$ indicates that the variable X_j plays a significant role in defining the direction of Z_i. These "weights" provide what the component **is**.

Theory of PCA (5)

If $\mathbf{X}$ follows a multivariate normal distribution $\mathcal{N}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then:

$$(\mathbf{x} - \boldsymbol{\mu})^t \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2$$

We have ellipsoids with axis $\pm c\sqrt{\lambda_i}\mathbf{e}_i$ with $i = 1, \ldots, p$. A point in the ith axis will have coordinates $\alpha \mathbf{e}_i^t = (e_{i1}, e_{i2}, \ldots, e_{ip})$ in the coordinate system that has origin $\boldsymbol{\mu}$ and axes that are parallel to the original $x_1, \ldots, x_p$ coordinates.

Centering $\mathbf{X}$ we get:

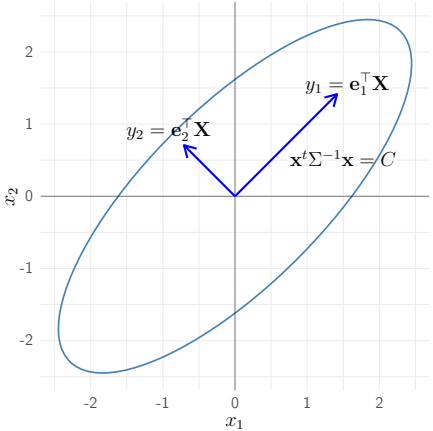
$$c^2 = \mathbf{x}^t \boldsymbol{\Sigma}^{-1} \mathbf{x} = \frac{1}{\lambda_1}(\mathbf{e}_1^t \mathbf{x})^2 + \cdots + \frac{1}{\lambda_p}(\mathbf{e}_p^t \mathbf{x})^2$$

If $z_i = \mathbf{e}_i^t \mathbf{x}$ then:

$$c^2 = \frac{1}{\lambda_1} y_1^2 + \cdots + \frac{1}{\lambda_p} y_p^2$$

Theory of PCA (6)

We got the ellipsoid equations of axis $\mathbf{e}_1, \ldots, \mathbf{e}_2$. The length of each axis is the corresponding eigenvalue: $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$.



Elements of PCA: Notation and definitions (1)

- **Principal Components Scores (Z_i; $\boldsymbol{Z}$)**: Uncorrelated random variables formed by the linear combination of original variables. Geometrically, they represent the projection of the data onto the directions of maximum variance.
- **Loadings ($\mathbf{e}_i$; $\boldsymbol{V}$)**: The eigenvectors of the covariance (or correlation) matrix. They define the orientation of the new axes; their magnitude and sign indicate the strength and direction of the relationship between the original variables and the components.
- **Scaled loadings**: Eigenvectors scaled by the square root of their corresponding eigenvalues $\mathbf{e}_i * \sqrt{\lambda_i}$. These represent the covariance between the original variables and the principal components.

Elements of PCA: Notation and definitions (2)

Note

Observe that $Var(Z_i) = \frac{1}{n-1} < Z_i^t, Z_i >= \frac{1}{n-1}||Z_i||^2 \implies ||Z_i|| = std(Z_i) * \sqrt{n-1}$ then the **scaled loadings** are defined as $l_i = \sqrt{n-1}\frac{Cov(X_j,X_i)}{||Z_i||} = \frac{Cov(X_j,Z_i)}{std(Z_i)} = Cov(X_j, Z_{std_i})$ where Z_{std_i} are the standardized principal components scores.

In matrix notation:

$$L = VD_\lambda^{\frac{1}{2}}$$

Sample principal components (1)

In practice, we observe n independent samples from a p-dimensional population with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. We estimate these using the sample mean $\overline{\boldsymbol{x}}$, covariance matrix $\mathbf{S}$, and sample correlation $\mathbf{R}$. Consistent with the previous approach, we aim to obtain uncorrelated linear combinations of the original variables that exhibit maximum sample variance.

Sample principal components (2)

Let $\mathbf{S}$ be the $(p \times p)$ sample covariance matrix with eigenvalue-eigenvector pairs $(\hat{\lambda}_1, \hat{\mathbf{e}}_1), (\hat{\lambda}_2, \hat{\mathbf{e}}_2) \ldots (\hat{\lambda}_p, \hat{\mathbf{e}}_p)$ where $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p$. The i-th principal component is given by

$$\hat{\mathbf{z}}_i = \hat{\mathbf{e}}_i^t \mathbf{X} = \hat{e}_{i1} X_+ \hat{e}_{i2} X_2 + \cdots + \hat{e}_{ip} X_p \qquad i = 1, 2, \ldots, p$$

where **x** is any observation on the variables $X_1, X_2, \ldots, X_p$. Also

- Sample variance$(\hat{\mathbf{z}}_k) = \hat{\lambda}_k, \qquad i = 1, 2, \ldots, p$
- Sample covariance$(\hat{\mathbf{z}}_i, \hat{\mathbf{z}}_k) = 0, \qquad i \neq k$
- Total sample variance $= \sum_{i=1}^{p} s_{ii} = \hat{\lambda}_1 + \hat{\lambda}_2 + \cdots + \hat{\lambda}_p$
- $r_{z_i x_j} = \hat{e}_{ik} \frac{\sqrt{\hat{\lambda}_i}}{\sqrt{s_{kk}}}$

Sample principal components (3)

In sample principal component analysis, the scores $\hat{\mathbf{z}}_i = \hat{\mathbf{e}}_i(\mathbf{x} - \overline{\boldsymbol{x}})$ are realizations of the theoretical components $\mathbf{Z}_i = \mathbf{e}_i^t(\mathbf{X} - \boldsymbol{\mu})$. For a multivariate normal population, the vector of components $\mathbf{Z}$ follows a distribution $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Lambda})$ where $\boldsymbol{\Lambda} = \mathbf{diag}(\lambda_1, \ldots \lambda_\mathbf{p})$.

Using a sample of observations $\mathbf{x}_j$, we estimate the population parameters $\boldsymbol{\mu}$ with $\overline{\boldsymbol{x}}$ and $\boldsymbol{\Sigma}$ with $\mathbf{S}$.

If $\mathbf{S}$ is positive definite, the quadratic form

$$(\mathbf{x} - \overline{\boldsymbol{x}})^t \mathbf{S}^{-1} (\mathbf{x} - \overline{\boldsymbol{x}}) = c^2$$

defines an ellipsoid that estimates the constant density contour of the population distribution, $(\mathbf{x} - \boldsymbol{\mu})^t \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2$. These estimated contours can be superimposed on a scatterplot to visualize the data's structure and variability.

Sample principal components (4)

We can express the data in the new coordinates (contour axes) getting:

- Data centered at $\overline{\boldsymbol{x}}$ whose axes are the eigenvectors of $\mathbf{S}^{-1}$.
- Lengths of the axes are proportional to $\sqrt{\hat{\lambda}_i} \quad i = 1, \ldots, p$ with $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p$, the eigenvalues of $\mathbf{S}$.
- $\|\mathbf{e}_i\| = 1$ then $|\hat{\mathbf{z}}_i| = \hat{\mathbf{e}}_i^t(\mathbf{x} - \overline{\boldsymbol{x}})$ gives the length of the projection of the vector $(\mathbf{x} - \overline{\boldsymbol{x}})$

Sample principal components (5)

Since the **population** covariance is **unknown**, we use the sample covariance matrix **S** to estimate $\mathbf{\Sigma}$, resulting in the sample decomposition:

$$\mathbf{S} = \hat{\mathbf{V}}\mathbf{D}_l\hat{\mathbf{V}}^t$$

Important

Consequently, the principal component scores are estimated using the sample loadings:

$$\hat{\mathbf{Z}}_{n\times p} = \mathbf{X}_{n\times p}\hat{\mathbf{V}}_{p\times p}$$

Sample principal components (6)

Steps to obtain the sample PCA solution:

1. The data, $\mathbf{X}_{n\times p}$, needs to be centered by performing a translation. This ensures that all observed random variables have a zero mean. The centered data matrix is obtained, denoted $\mathbf{X}_c$
2. With the centered data, now we compute the correlation matrix $\mathbf{S} = \frac{1}{(n-1)}\mathbf{X}_c^t\mathbf{X}_c$.
3. Compute SVD or eigendecomposition on $\mathbf{S} = \hat{\mathbf{V}}\mathbf{D}_l\hat{\mathbf{V}}^t$.
4. Project the centered data onto the directions of maximum variability to obtain the principal component scores:

$$\hat{\mathbf{Z}} = \mathbf{X}_c\hat{\mathbf{V}}$$

Standardized sample principal components (1)

If **size effects** are present due to large variances in some variables, we need to normalize the variances. This is accomplished modifying a bit the previous steps:

1. Center and standardize the data matrix: $\mathbf{X}_s = \frac{\mathbf{X}-\bar{\mathbf{x}}}{\sigma}$
2. Compute the correlation matrix: $\mathbf{R} = \frac{1}{(n-1)}\mathbf{X}_s^t\mathbf{X}_s$.
3. Compute the SVD or eigendecomposition: $\mathbf{R} = \hat{\mathbf{V}}\mathbf{D}_l\hat{\mathbf{V}}^t$.
4. Project the scaled data on the directions of maximum variability to obtain the principal component scores:

$$\hat{\mathbf{Z}} = \mathbf{X}_s\hat{\mathbf{V}}$$

Standardized sample principal components (2)

- Consider the weight matrix $\mathbf{D}^{-\frac{1}{2}}$, then:

$$\mathbf{z}_j = \mathbf{D}^{-\frac{1}{2}}(\mathbf{x}_j - \overline{x}_1) = \begin{bmatrix} \frac{x_{j1}-\overline{x}_1}{\sqrt{s_{11}}} \\ \vdots \\ \frac{x_{jp}-\overline{x}_p}{\sqrt{s_{pp}}} \end{bmatrix}$$

- Applying the weights to all the individuals we get the standardized sample matrix $\mathbf{Z}_{n\times p}$

- $\overline{\mathbf{z}} = \frac{1}{n}(\mathbf{1}^t\mathbf{z}^t) = \frac{1}{n}\mathbf{z}^t\mathbf{1} = 0$

- $\mathbf{S}_z = \frac{1}{n-1}(\mathbf{Z} - \frac{1}{n}\mathbf{1}^t\mathbf{1}\mathbf{Z})^t(\mathbf{Z} - \frac{1}{n}\mathbf{1}^t\mathbf{1}\mathbf{Z}) = \frac{1}{n-1}(\mathbf{Z} - \mathbf{1}\overline{\mathbf{z}}^t)^t(\mathbf{Z} - \mathbf{1}\overline{\mathbf{z}}^t) = \frac{1}{n-1}\mathbf{Z}^t\mathbf{Z} = \mathbf{R}$

Standardized sample principal components (3)

If $\mathbf{z}_1, \ldots, \mathbf{z}_n$ are standardized observations with covariance matrix $\mathbf{R}$, the ith sample principal component is

$$\hat{\mathbf{z}} = \hat{\mathbf{e}}_i^t \mathbf{z} = \hat{e}_{i1} z_1 + \hat{e}_{i2} z_2 + \cdots + \hat{e}_{ip} z_p \qquad i = 1, 2, \ldots, p$$

where $(\hat{\lambda}_i, \hat{\boldsymbol{e}}_i)$ is the ith eigenvalue-eigenvector pair of $\mathbf{R}$ with $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p \geq 0$. Also

- Sample covariance of $\hat{\mathbf{z}}_i = \hat{\lambda}_i \qquad i = 1, 2, \ldots, p$
- Sample covariance of $(\hat{\mathbf{z}}_i, \hat{\lambda}_k) = 0 \qquad i \neq k$
- Total (standardized) sample variance equals $\mathrm{tr}(\mathbf{R}) = p = \hat{\lambda}_1 + \hat{\lambda}_2 + \cdots + \hat{\lambda}_p$
- and $r_{\hat{y}_i z_k} = \hat{e}_{ik} \sqrt{\hat{\lambda}_i} \qquad i, k = 1, 2, \ldots, p$

Standardized sample principal components (3)

It is easy to verify that

- Proportion of (standardized) sample variance due to the *i*th sample principal component is:

$$\frac{\hat{\lambda}_i}{p} \qquad i = 1, 2, \ldots, p$$

NOTE: As a rule of thumb some authors suggest to retain those components whose variances $\hat{\lambda}_i$ are greater than unity or those with a proportion $\frac{1}{p}$ of the total variance

Interpretations in PCA (1)

We can interpret PCA through:

- The loadings.
- The correlations between observed variables and the scores.
- The biplot.

We can also interpret variables and observations through:

- Variables: The scaled loadings.
- Observations: The principal component scores.

Interpretations in PCA (2)

Interpretations using the loadings coefficients:

- **Loading sign**: Direction of the relation between variable i and PC j.
- **Loading magnitude**: How much variable i variance contributed to the PC j formation.

When a group of variables have high coefficient values in a loading, we can interpret a common theme that is reflected by the principal component.

Interpretations in PCA (3)

Interpretations based on covariances between the original variables and the principal components:

- If $\mathbf{X}$ centered but **not** standardized then $Cov(\mathbf{X}, \mathbf{Z}) = \tilde{\mathbf{L}}$. $\tilde{\mathbf{L}} = \mathbf{V}\mathbf{D}_\lambda$ are the **full scaled loadings**.
- If $\mathbf{X}$ centered but **not** standardized then $Cov(\mathbf{X}, \mathbf{Z}_{std}) = \mathbf{L}$, where $\mathbf{Z}_{std} = \mathbf{Z}\mathbf{D}_\lambda^{-\frac{1}{2}}$ is the matrix of **standardized principal scores** and $\mathbf{L} = \mathbf{V}\mathbf{D}_\lambda^{\frac{1}{2}}$ is the matrix of **scaled loadings**.
- If $\mathbf{X}$ centered and standardized then $Cov(\mathbf{X}, \mathbf{Z}) = \tilde{\mathbf{L}}$.
- If $\mathbf{X}$ centered and standardized then $Cov(\mathbf{X}, \mathbf{Z}_{std}) = \mathbf{L}$.

Interpretations in PCA (3b)

Interpretations based on correlations between the original variables and the principal components:

- If $\mathbf{X}$ centered but **not** standardized then $Cor(\mathbf{X}, \mathbf{Z}) = \mathbf{S}^{-1}\mathbf{L}$. Where $\mathbf{S}^{-1}$ is the diagonal matrix of standard deviations from the original variables.
- If $\mathbf{X}$ centered and standardized then $Cor(\mathbf{X}, \mathbf{Z}) = \mathbf{L}$.

Note: If $\mathbf{X}$ centered and standardized then $Cor(\mathbf{X}, \mathbf{Z}_{std}) = Cov(\mathbf{X}, \mathbf{Z}_{std})$

Interpretations in PCA (4)

Interpretations using the scaled loadings:

- **Contributions** of each observed variable to the formation of a principal component.
- **Squared cosines** of an observed variable with each one of the principal components.

Further information regarding contributions and squared cosines will be provided in a subsequent slide.

Interpretations in PCA (5)

Interpretations using the principal component scores:

- **Contributions** of each observation to the variance of a principal component.
- **Squared cosines** of an observation with each one of the principal components.

Further information regarding contributions and squared cosines will be provided in a subsequent slide.

Interpretations in PCA (6)

When two principal components capture a large amount of variance (more than the 80% of the total variance) then it is possible to get interpretations using visualizations:

- Scatterplot of the principal component scores help to understand the individuals factor space and interpret the relations between the principal components (axis) with the individuals and relations between the individuals.
- Plot of the variables. Variables represented as vectors in the variables factor space help to understand the relationships between the principal components (axis) and the variables and the relations between the variables.
- The biplot summarizes the information from earlier plots and reveals relationships between individuals and variables through individual projections onto variable axes.

Squared cosines - variables

- **Squared cosines**: Measures quality of representation in the factorial space with respect to the original space by measuring the cosine between each one of the variables i with each one of the factorial axes using the squared scaled loadings l_{ij}^2.

$$cos^2(i,j) = \frac{l_{i,j}^2}{\sum_{j=1}^{p} l_{i,j}^2}$$

Observe that the row marginal is $\sum_{j=1}^{p} l_{ij}^2 = Var(X_j)$. Squared cosines have range from 0 to 1 and the sum of all cos^2 for each variable equals 1. A good representation happens when with few principal components the sum is nearly 1.

Contributions - variables

- **Contributions**: Proportion of the variance of each observed variable that contributed to each one of the principal components formation using the squared scaled loadings l_{ij}^2.

$$ctr(i,j) = \frac{l_{ij}^2}{\sum_{i=1}^{n} l_{ij}^2} * 100$$

Observe that the column marginal is $\sum_{i=1}^{n} l_{ij}^2 = \lambda_j$. So the sum up of all the contributions is 100%. Contributions help to characterize size effects: When a contribution of a variable is way larger than the contributions of the others.

Squared cosines - individuals

- **Squared cosines**: Measures quality of representation in the factorial space with respect to the original space by measuring the cosine between each one of the individuals i with each one of the factorial axes. The squared principal component scores are used in the calculations, z_{ij}^2.

$$cos^2(i,j) = \frac{z_{i,j}^2}{\sum_{j=1}^{p} z_{i,j}^2}$$

In this case no meaning is applied to the row marginal. Squared cosines have range from 0 to 1 and the sum of all cos^2 for each observation equals 1. A good representation happens when with few principal components the sum is nearly 1.

Contributions - individuals

- **Contributions**: Proportion of the contribution of each observation (individual) to each one of the principal components formation. The squared principal component scores are used in the calculations, z_{ij}^2.

$$ctr(i,j) = \frac{z_{i,j}^2}{\sum_{i=1}^{n} z_{i,j}^2} * 100$$

Observe that the columns marginals, $\sum_{i=1}^{n} z_{i,j}^2$, are equal to the sum of the squares used to compute the variance of the principal components: $\sum_{i=1}^{n} z_{i,j}^2 = D_\lambda(n-1)$. The sum up of all the contributions is 100%. Contributions help to characterize observations that are **outliers** as a high contribution value means that the observation is far from the origin (the mean).

Computing PCA solution using SVD (1)

Consider a matrix $\mathbf{X}_t$ as:

$$\mathbf{X}_t = \frac{1}{\sqrt{n-1}}\mathbf{X}_c$$

This matrix is related through the covariance matrix $\mathbf{S}$ in the following way:

$$\mathbf{X}_t^t\mathbf{X}_t = \mathbf{S} = \mathbf{V}\mathbf{D}_\lambda\mathbf{V}^t$$

Then $\mathbf{X}_t$ can be decomposed (by SVD) as:

$$\mathbf{X}_t = \mathbf{U}\mathbf{D}_s\mathbf{V}^t$$

Now consider the PC scores matrix $\mathbf{Z}$. We know that:

$$\mathbf{Z} = \mathbf{X}_c\mathbf{V}$$

It turns out that it is equivalent to:

$$\mathbf{Z} = \mathbf{X}_c\mathbf{V} = \sqrt{n-1}\mathbf{X}_t\mathbf{V} = \sqrt{n-1}\mathbf{U}\mathbf{D}_s\mathbf{V}^t\mathbf{V} = \sqrt{n-1}\mathbf{U}\mathbf{D}_s$$

$$\mathbf{D}_s = \mathbf{D}_\lambda^{\frac{1}{2}} \qquad \mathbf{Z}_s = \mathbf{Z}\mathbf{D}_s^{-1} = \sqrt{n-1}\mathbf{U}$$

Computing PCA solution using SVD (2)

and we have

$$\mathbf{X}_c = \sqrt{n-1}\mathbf{X}_t = \sqrt{n-1}\mathbf{U}\mathbf{D}_s\mathbf{V}^t = \mathbf{Z}\mathbf{V}^t$$

Note that:

- The variables are also linear combinations of the components.
- PCA provides a low rank approximation to the centered data matrix that is optimal in the least squares sense.
- PCA **also** provides a low rank approximation to the covariance matrix that is optimal in the least squares sense.

PCA Biplots

- For each formulation of the PCA ($\mathbf{S}$ or $\boldsymbol{R}$) and the imputation of the singular values (left or right), the corresponding biplots will have different properties.

- Remember the notation:
  - $\mathbf{F}$: Row coordinates.
  - $\mathbf{G}$: Column coordinates.
  - p suffix indicates principal coordinates (uses the singular values).
  - s suffix indicates standard coordinates (not uses the singular values).

- The available options are then: $\mathbf{F}_s$ with $\mathbf{G}_p$ or $\mathbf{F}_p$ with $\mathbf{G}_s$

$$\mathbf{F}_p = \mathbf{U}\mathbf{D}_s \qquad \mathbf{F}_s = \mathbf{U}$$

$$\mathbf{G}_s = \mathbf{V} \qquad \mathbf{G}_p = \mathbf{V}\mathbf{D}_s$$

Example (1)

We have 103 successive weeks observations of weekly rates return for five stocks: JP Morgan, Citibank, Wells Fargo, Royal Dutch Shell and ExxonMobil. Let $x_1, x_2, \ldots, x_5$ denote observed weekly rates returns. Then

$$\overline{\boldsymbol{x}}^t = \begin{bmatrix} .0011 & .0007 & .0016 & .0040 & .0040 \end{bmatrix}$$

and

$$\mathsf{R} = \begin{pmatrix} 1.000 & .632 & .511 & .115 & .155 \\ .632 & 1.000 & .573 & .322 & .213 \\ .511 & .573 & 1.000 & .183 & .146 \\ .115 & .322 & .183 & 1.000 & .683 \\ .155 & .213 & .146 & .683 & 1.000 \end{pmatrix}$$

Example (2)

The obtained eigenvalues and normalized eigenvectors of **R** are:

$$\hat{\lambda}_1 = 2.437, \qquad \hat{\boldsymbol{e}}_1^t = \begin{bmatrix} .469, & .532, & .465, & .387, & .361 \end{bmatrix}$$

$$\hat{\lambda}_2 = 1.407, \qquad \hat{\boldsymbol{e}}_2^t = \begin{bmatrix} -.368, & -.236, & -.315, & .585, & .606 \end{bmatrix}$$

$$\hat{\lambda}_3 = .501, \qquad \hat{\boldsymbol{e}}_3^t = \begin{bmatrix} -.604, & -.136, & .772, & .0.93, & -.109 \end{bmatrix}$$

$$\hat{\lambda}_4 = .400, \qquad \hat{\boldsymbol{e}}_4^t = \begin{bmatrix} .363, & -.629, & .289, & -.381, & .493 \end{bmatrix}$$

$$\hat{\lambda}_5 = .255, \qquad \hat{\boldsymbol{e}}_5^t = \begin{bmatrix} .384, & -.496, & .071, & .595, & -.498 \end{bmatrix}$$

Example (3)

The two first sample principal components are:

$$\hat{\mathbf{y}}_1 = \hat{\mathbf{e}}_1^t \mathbf{z} = .469z_1 + .532z_2 + .465z_3 + .387z_4 + .361z_5$$
$$\hat{\mathbf{y}}_2 = \hat{\mathbf{e}}_2^t \mathbf{z} = -.368z_1 - .236z_2 - .315z_3 + .585z_4 + .606z_5$$

These components account for:

$$(\frac{\hat{\lambda}_1 + \hat{\lambda}_2}{p})100\% = (\frac{2.437 + 1.407}{5})100\% = 77\%$$

The first component can be interpreted as an equally weighted sum (or index) of the five stocks, like a general stock market component. The second represents a contrast between type of stocks: banks (variables x_1, x_2 and x_3) vs oil industry (x_4 and x_5).

Exercise: Size effect

We have the following covariance and correlation matrices from the random variables $\{\mathbf{X_1}, \mathbf{X_2}\}$.

$$\boldsymbol{\Sigma_X} = \begin{bmatrix} 1 & 4 \\ 4 & 100 \end{bmatrix}$$

$$\boldsymbol{\rho_X} = \begin{bmatrix} 1 & .4 \\ .4 & 1 \end{bmatrix}$$

Check the effect of the variance when performing PCA on one or to the other. Also check the relative importance of the variables.

PCA Biplots: Form biplot

- From the SVD we have $\mathbf{X}_c = \sqrt{n-1}\mathbf{U}\mathbf{D}_s\mathbf{V}^t = \mathbf{F}_p\mathbf{G}_s^t$
- Take $\mathbf{F}_p = \sqrt{n-1}\mathbf{U}\mathbf{D}_s$ and $\mathbf{G}_s = \mathbf{V}$
- A biplot of the centered data matrix is obtained by plotting the two principal components (row points) and the rows of the first two eigenvectors (as arrows).
- Because $\mathbf{X}_c\mathbf{X}_c^t = (n-1)\mathbf{U}\mathbf{D}_\lambda\mathbf{U}^t = \mathbf{F}_p\mathbf{F}_p^t$ then the squared Euclidean distance between rows of $\mathbf{X}_c$ equals the squared Euclidean distance between rows of $\mathbf{F}_p$. This means that the Euclidean distance in the biplot is preserved.

PCA Biplots: Covariance biplot (1)

- From the SVD we have $\mathbf{X}_c = \sqrt{n-1}\mathbf{U}\mathbf{D}_s\mathbf{V}^t = \mathbf{F}_s\mathbf{G}_p^t$
- Take $\mathbf{F}_s = \sqrt{n-1}\mathbf{U}$ and $\mathbf{G}_s = \mathbf{V}\mathbf{D}_s$
- $\mathbf{F}_s$ contains standardized principal components and $\mathbf{G}_p$ contains covariances between variables and components.
- A biplot of the centered data matrix is obtained by plotting the two standardized components (row points) and the covariances of the first two components with the variables (as arrows).
- Observe that $\mathbf{X}_c\mathbf{S}^{-1}\mathbf{X}_c^t = (n-1)\mathbf{U}\mathbf{U}^t = \mathbf{F}_s\mathbf{F}_s^t$ implies that the squared Mahalanobis distance between rows of $\mathbf{X}_c$ equals the squared Euclidean distance between rows of $\mathbf{F}_s$, i.e.: $(\mathbf{x}_i - \mathbf{x}_j)^t\mathbf{S}^{-1}(\mathbf{x}_i - \mathbf{x}_j) = (\mathbf{f}_i - \mathbf{f}_j)^t(\mathbf{f}_i - \mathbf{f}_j)$. This means that the Euclidean distances in the biplot represent Mahalanobis distances.

PCA Biplots: Covariance biplot (2)

- $\mathbf{G}_p\mathbf{G}_p^t = \mathbf{V}\mathbf{D}_s^2\mathbf{V}^t = \mathbf{S} = \frac{1}{n-1}\mathbf{X}_c^t\mathbf{X}_c$
- $cos(\mathbf{g}_i, \mathbf{g}_j) = \frac{\langle \mathbf{g}_i, \mathbf{g}_j \rangle}{\|\mathbf{g}_i\|\|\mathbf{g}_j\|} = \frac{\frac{1}{n-1}\langle \mathbf{x}_i^t, \mathbf{x}_j \rangle}{\frac{1}{\sqrt{n-1}}\|\mathbf{x}_i\|\frac{1}{\sqrt{n-1}}\|\mathbf{x}_j\|} = \frac{\frac{1}{n-1}\langle \mathbf{x}_i^t, \mathbf{x}_j \rangle}{\sqrt{\frac{1}{n-1}\langle \mathbf{x}_i^t, \mathbf{x}_i \rangle}\sqrt{\frac{1}{n-1}\langle \mathbf{x}_j^t, \mathbf{x}_j \rangle}} = r(\mathbf{x}_i, \mathbf{x}_j)$
- $\|\mathbf{g}_i\| = \sqrt{(1/(n-1))\langle \mathbf{x}_i^t, \mathbf{x}_i \rangle} = s_{x_i}.$
- For standardized data: $\|\mathbf{g}_i\| = \sqrt{r^2\langle \mathbf{x}_i, \mathbf{F}_1 \rangle + r^2\langle \mathbf{x}_i, \mathbf{F}_2 \rangle + \cdots + r^2\langle \mathbf{x}_i, \mathbf{F}_p \rangle} = \sqrt{R^2}$

Categorical variables

- Each value of the categorical variable splits individuals into **groups**, having as many groups as levels has the categorical variable.
- For each group we can compute the **centroid** as the average point or center of gravity.
- Then we **project the centroid** onto the rows space.
- A plot of this projection can give us **extra information** about **regions** of the cloud of row points and also information about how to interpret the **directions** of the Principal Components.

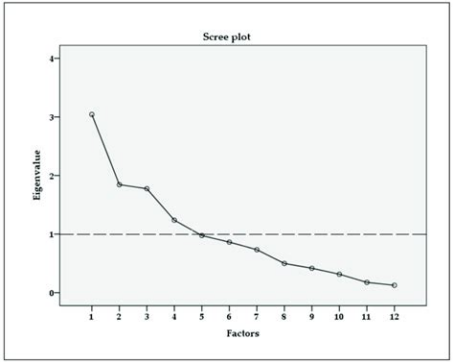
Active variables vs Supplementary variables

- PC axis are built using a set of variables. These variables are called **active variables**.
- It is possible to use other variables that have not been used in the construction of the PCA axis. Just projecting those into the PC. These variables are called **supplementary variables**
- The **position** of a supplementary variable in the plane built by the PCs allows us to visualize the **relationship** of the variable **with** the set of **active variables** through the PC.
- A high correlation between a supplementary variable and one of the PC indicates a high correlation between that variable and those active variables that have contributed the most in the construction of the used PC.

The number of principal components (1)

There exists different heuristics to select the number of principal components:

- *Scree plot*: Plot of the **eigenvalues ordered from largest to smallest** where the y-axis is the magnitude of each eigenvalue and the x-axis is its number.



The number of principal components (2)

- *Kaiser rule*: Take as number of components those **eigenvalues above the average** (it will be 1 if the data is standardized). If original variables are uncorrelated then their variances and the eigenvalues.
- *Cumulative percentage of variance*: Components up to a fixed percentage (e.g. 80%) or information desired to be retained.
- *Elbow rule*: Take as number of components those **before** the Scree plot makes an "elbow" or the point in which **the curve flattens** significantly.
- *Statistical test*: There are statistical test of significance for testing the importance of the eigenvalues λ_i. E.g.: Create a shuffled version of the original data and calculate their eigenvalues, then keep only the eigenvalues higher than those generated by the random data.

Best k rank approximation

The **optimal** rank-k approximation under the **least-squares criterion** is given by:

$$\hat{\mathbf{X}} = Z_{[:,1:k]} V^t_{[1:k,1:k]}$$

where $k = 1, \ldots, p$

PCA in R

Some libraries and methods in R to calculate PCA:

- *prcomp*: Method from base R that computes the principal component scores and the square root of the variances captured.
- *FactoMineR*: Library with methods to compute all the results possible from PCA and also provides beautiful representations for the biplots (*factoextra*).

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